

```
# Creating a list
fruits = ['apple', 'banana', 'cherry']
```

```
# Adding elements
fruits.append('orange')
fruits.insert(1, 'grapes')
```

```
# Slicing
print(fruits[1:3]) # ['grapes', 'cherry']
```

```
➞ ['grapes', 'banana']
```

```
# Looping through list
for fruit in fruits:
    print(fruit)
```

```
➞ apple
    grapes
    banana
    cherry
    orange
```

```
import numpy as np
```

```
# Creating an array
arr = np.array([10, 20, 30, 40])
```

```
# Basic operations
print(arr + 5)      # [15 25 35 45]
print(arr * 2)      # [20 40 60 80]
```

```
➞ [15 25 35 45]
   [20 40 60 80]
```

```
# Indexing and slicing
print(arr[1:3])     # [20 30]
```

```
➞ [20 30]
```

```
# 2D Array
matrix = np.array([[1, 2], [3, 4]])
print(matrix)
```

```
➞ [[1 2]
   [3 4]]
```

```
import pandas as pd
```

```
# Creating a DataFrame
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Marks': [85, 90, 95]
}
df = pd.DataFrame(data)
```

```
# Viewing data
print(df.head())
```

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⇒
```

	Name	Marks
0	Alice	85
1	Bob	90
2	Charlie	95

```
# Adding a new column
df['Grade'] = ['A', 'A+', 'A+']
```

```
# Filtering rows
print(df[df['Marks'] > 85])
```

```
⇒
```

	Name	Marks	Grade
1	Bob	90	A+
2	Charlie	95	A+

```
# Creating a dictionary
student = {
    'name': 'Alice',
    'age': 21,
    'course': 'Computer Science'
}
```

```
# Accessing and updating
print(student['name'])    # Output: Alice
student['age'] = 22       # Updating age
```

```
⇒ Alice
```

```
# Adding new key
student['college'] = 'MGM COE'
```

```
# Loop through dictionary
for key, value in student.items():
    print(f"{key}: {value}")
```

```
⇒ name: Alice
   age: 22
   course: Computer Science
   college: MGM COE
```